

# Supporting Information for BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Biological Prior Knowledge

This document reports values recoverable from the released Davis records, configuration files, saved result files, source code, and the fixed BindingDB external-validation manifest included with this revision. “Classifier score” denotes the sigmoid output for the binary high-affinity class; it is not an experimental affinity. Model-native attention and Grad-CAM values are attribution scores, not contact assignments.

## Model configuration and input features

Table S1: Model architecture confirmed from the checkpoint-embedded model configurations.

| Component              | Archived setting                                                  |
|------------------------|-------------------------------------------------------------------|
| Drug encoder           | GINE; 3 layers; hidden dimension 256; dropout 0.20                |
| Atom and bond input    | 52 node features; 16 edge features                                |
| Protein language model | <code>esm2_t30_150M_UR50D</code> ; 640-dimensional residues       |
| Protein projection     | 256-dimensional projection; 32-dimensional domain-label embedding |
| Cross-attention        | 8 heads; attention dimension 256; dropout 0.10                    |
| Pooling                | Gated                                                             |
| Prediction head        | Hidden dimensions 256 and 128; dropout 0.30; classification task  |

Table S2: Protein-residue input channels and their availability.

| Channel                 | Archived implementation and limitation                                                                                                                                                                            |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ESM-2                   | Frozen 640-dimensional residue embeddings; proteins are truncated at 1,200 residues.                                                                                                                              |
| Physicochemical vector  | Four values per residue: hydrophobicity, normalized molecular weight, charge, and polarity.                                                                                                                       |
| Domain-label channel    | The configuration reserves a 32-dimensional embedding. The label builder uses <code>NONE</code> when no per-protein JSON annotation is available.                                                                 |
| Annotation availability | No <code>data/domain_annotations/*.json</code> files are present in the released package. Thus the reproducible Davis input uses the unknown label at every residue and is not curated Pfam/InterPro information. |

Table S3: Training provenance for the historical entity-split checkpoints and the separately trained exact-sequence-grouped strict models.

| Setting                              | Value                                                                                                                                                                                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Historical entity-split scope        | Model architecture only; the three historical entity-split checkpoints do not retain optimizer, scheduler, warmup, or loss state                                                                                           |
| Historical Random and Drug-ID logs   | Final-run headers record weighted BCE, label smoothing 0.05, AdamW ( $10^{-4}$ , $10^{-5}$ ), and cosine warmup                                                                                                            |
| Historical Target-ID trace           | Older <code>run_split.py</code> header; only loss and validation AUROC are logged, so optimizer, warmup, and smoothing are not recoverable from the artifact                                                               |
| Historical threshold protocol        | A validation F1 threshold is frozen before each canonical test evaluation                                                                                                                                                  |
| Strict-model initialisation and data | Each exact-sequence-grouped model was newly initialised with seed 42 and trained only on its new Davis training partition, using the same preprocessing and architecture                                                   |
| Strict-model optimisation            | Weighted BCE with 0.05 label smoothing; AdamW (learning rate $10^{-4}$ , weight decay $10^{-5}$ ); batch size 64; gradient-norm clipping 1.0; five-epoch linear warm-up followed by cosine decay; maximum 100 epochs       |
| Strict-model selection               | Highest validation AUROC with patience 15; validation F1 threshold selected after loading that checkpoint and frozen for test evaluation                                                                                   |
| Strict cold-target artifact          | <code>revision/analysis/sequence_grouped_cold_target.pt</code> ; best epoch 24 of 39; validation AUROC 0.911568; threshold 0.837581; SHA-256 <code>8ea58e1f056b5e33c309787880a811c797393510c459d9442b5dbe67e632fe5c</code> |
| Strict cold-both artifact            | <code>revision/analysis/sequence_grouped_cold_both.pt</code> ; best epoch 6 of 21; validation AUROC 0.910630; threshold 0.935704; SHA-256 <code>8b7c0f374983a35163a7d3243c89732504a5a97ef4d10c557b2841a627873972</code>    |

Table S4: Computational environment documented during final reproducibility audit.

| Item                           | Value                                                                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Python                         | 3.10.14                                                                                                                                              |
| PyTorch                        | 2.4.0+cu121                                                                                                                                          |
| PyTorch Geometric              | 2.6.1                                                                                                                                                |
| RDKit                          | 2025.3.6                                                                                                                                             |
| <code>fair-esm</code>          | 2.0.0                                                                                                                                                |
| Biopython and RapidFuzz        | 1.87 and 3.14.5                                                                                                                                      |
| CUDA availability during audit | Available                                                                                                                                            |
| Audit GPU                      | NVIDIA GeForce RTX 4080                                                                                                                              |
| Strict-split run record        | <code>device: cuda</code> in <code>strict_split_metrics.json</code>                                                                                  |
| Historical checkpoint files    | <code>best.pt</code> , <code>best_random.pt</code> , <code>best_cold_target.pt</code> , <code>best_cold_drug.pt</code>                               |
| Strict-model checkpoint files  | <code>revision/analysis/sequence_grouped_cold_target.pt</code> and <code>revision/analysis/sequence_grouped_cold_both.pt</code> (hashes in Table S3) |

## Dataset, partitions, and metrics

Table S5: Canonical current-release Davis test-partition summaries. Values are from the full-precision checkpoint reconstruction, frozen validation thresholds, and the associated prediction files.

| Partition          | Test pairs | Positives | AUROC | AUPRC | F1    | Precision | Recall |
|--------------------|------------|-----------|-------|-------|-------|-----------|--------|
| Random             | 6,011      | 276       | 0.904 | 0.560 | 0.565 | 0.602     | 0.533  |
| Target-ID-held-out | 5,984      | 250       | 0.930 | 0.525 | 0.534 | 0.517     | 0.552  |
| Drug-ID-held-out   | 5,746      | 245       | 0.733 | 0.167 | 0.100 | 0.273     | 0.061  |

Table S6: Bootstrap intervals for the canonical current-release entity-based partitions. Each interval uses 600 ordinary pair-level bootstrap replicates sampled with replacement at the test-pair level, NumPy seed 42, and the 2.5th and 97.5th percentiles of valid replicate metrics. Replicates containing only one class are discarded. The main figure shows point estimates only.

| Partition          | AUROC 95% interval | AUPRC 95% interval |
|--------------------|--------------------|--------------------|
| Random             | [0.881, 0.924]     | [0.498, 0.624]     |
| Target-ID-held-out | [0.911, 0.948]     | [0.454, 0.584]     |
| Drug-ID-held-out   | [0.703, 0.763]     | [0.129, 0.211]     |

Table S7: Exact-sequence-grouped strict within-Davis evaluations from `strict_split_metrics.json`. The two models were trained from initialisation under the protocol in Table S3. Threshold-dependent quantities use the corresponding validation-set threshold; all strict-split entries are point estimates, not bootstrap confidence intervals.

| Pair accounting                    |             |                  |            |                |                            |  |
|------------------------------------|-------------|------------------|------------|----------------|----------------------------|--|
| Protocol                           | Train pairs | Validation pairs | Test pairs | Test positives | Excluded cross-block pairs |  |
| Exact-sequence-grouped cold-target | 21,080      | 2,992            | 5,984      | 208            | 0                          |  |
| Exact-sequence-grouped cold-both   | 15,190      | 264              | 1,144      | 38             | 13,458                     |  |

  

| Test metrics                       |       |       |           |       |           |        |  |
|------------------------------------|-------|-------|-----------|-------|-----------|--------|--|
| Protocol                           | AUROC | AUPRC | Threshold | F1    | Precision | Recall |  |
| Exact-sequence-grouped cold-target | 0.873 | 0.383 | 0.838     | 0.468 | 0.517     | 0.428  |  |
| Exact-sequence-grouped cold-both   | 0.552 | 0.038 | 0.936     | 0.000 | 0.000     | 0.000  |  |

For the cold-both protocol, drug identifiers and exact target-sequence groups are partitioned independently at 70/10/20. Only pairs in the corresponding train–train, validation–validation, and test–test blocks are retained; the remaining cross-block pairs are excluded. The cold-both validation block contains 10 positive pairs.

## Reproducibility details for revision analyses

**Identifier-partition similarity audits.** Drug similarities use RDKit 2025.3.6 Morgan fingerprints (radius 2, 1,024 bits, `includeChirality=False`) and Tanimoto similarity. Protein similarities for the original Target-ID-held-out audit use RapidFuzz 3.14.5 `fuzz.ratio/100`. For sequences  $x, y$ , this is the normalised Indel similarity  $1 - d_{\text{Indel}}/(|x| + |y|)$ , where substitutions are represented as one deletion plus one insertion.

**Strict-test global alignment.** Biopython 1.87 `Align.PairwiseAligner` is used in global mode with `match = 1`, `mismatch = 0`, `gap-opening = -1`, and `gap-extension = -0.5`. Full-length identity is the number of exact-residue matches divided by alignment length; the maximum across all training targets is retained for each held-out target.

**Bootstrap intervals.** The saved 95% intervals use ordinary, unstratified pair-level resampling with replacement, 600 requested replicates, and NumPy random seed 42. Each replicate contains the same number of test pairs as the corresponding test set; one-class replicates are skipped. Bounds are the 2.5th and 97.5th percentiles of the valid AUROC or AUPRC replicate distribution.

**Grad-CAM.** Hooks are attached to `drug_encoder.layers.2`, the third and final GINE message-passing layer. Gradients are taken from the pre-sigmoid positive-class logit, pooled across graph nodes to obtain channel weights, rectified, and then normalised separately within each molecular graph.

**Attention sparsity.** For pair  $q$ , residue scores  $s_{qj} = \sum_i M_{qij}$  are normalised as  $\tilde{s}_{qj} = s_{qj} / \max_k s_{qk}$ . The reported fraction is  $|\{(q, j) : \tilde{s}_{qj} < 0.1\}| / \sum_q L_q$ . It uses all 1,506 positive Davis pairs with the archived `checkpoints/best.pt`, not a single test set. Every pair is normalised before the 1,301,380 valid residue scores are concatenated; thus original training, validation, and test memberships are all included and longer targets contribute more residue-level observations. The archived flat-score array is `BioInteract/results/figure_data/attention_distribution.npz`; it is the sole source of the attention-distribution figure and cumulative curve in the main text.

**Residue-attribution summary.**

| Quantity                               | Value               |
|----------------------------------------|---------------------|
| Positive Davis pairs                   | 1,506               |
| Residue-score observations             | 1,301,380           |
| Mean / standard deviation              | 0.004351 / 0.046243 |
| Median                                 | 0.000272            |
| 95th / 99th percentile                 | 0.003617 / 0.057193 |
| Fraction below 0.1                     | 0.992319            |
| Attribution mass in top 1% / 5% scores | 0.788528 / 0.904465 |

**Exact-sequence-grouped target diagnostic controls.** All values use the 5,984-pair strict target test set with 208 positives.

| Control                              | AUROC | AUPRC |
|--------------------------------------|-------|-------|
| BioInteract                          | 0.873 | 0.383 |
| Drug-only Morgan logistic regression | 0.764 | 0.164 |
| Drug-promiscuity prior               | 0.764 | 0.163 |
| Target-only constant prior           | 0.500 | 0.035 |
| Label-shuffled drug-only control     | 0.407 | 0.029 |

## Feature and input audits

Table S8: Atom and bond feature specification from `src/data/mol_graph.py`.

| Level      | Feature family                                                   | Dimensions |
|------------|------------------------------------------------------------------|------------|
| Atom       | Atom type one-hot (20 choices including unknown)                 | 20         |
| Atom       | Hybridization, formal charge, and hydrogen-count one-hot vectors | 18         |
| Atom       | Degree, aromaticity, and ring membership                         | 3          |
| Atom       | Ring-size membership for sizes 3–8                               | 6          |
| Atom       | Donor/acceptor, Crippen logP and MR, Gasteiger charge            | 5          |
| Atom total |                                                                  | 52         |
| Bond       | Bond type, conjugation, ring membership, stereo, and direction   | 16         |

Table S9: Davis nominal ABL1 input audit. Every listed record has the same complete SHA-256 digest `cd6a78b82fede215cdb80a797f228098a41178fdde0753728f3781e6221fb1ee`; values in the last columns are the archived D0010 and D0017 affinities in nM. The nominal labels do not alter the model sequence input.

| Target ID | Nominal name | Length | Matches ABL1 | D0010 | D0017 |
|-----------|--------------|--------|--------------|-------|-------|
| T0001     | ABL1(E255K)  | 1167   | Yes          | 0.047 | 0.047 |
| T0002     | ABL1(F317I)  | 1167   | Yes          | 0.63  | 0.10  |
| T0003     | ABL1(F317I)p | 1167   | Yes          | 0.18  | 0.041 |
| T0004     | ABL1(F317L)  | 1167   | Yes          | 0.11  | 0.032 |
| T0005     | ABL1(F317L)p | 1167   | Yes          | 0.029 | 0.019 |
| T0006     | ABL1(H396P)  | 1167   | Yes          | 0.062 | 0.025 |
| T0007     | ABL1(H396P)p | 1167   | Yes          | 0.057 | 0.046 |
| T0008     | ABL1(M351T)  | 1167   | Yes          | 0.037 | 0.016 |
| T0009     | ABL1(Q252H)  | 1167   | Yes          | 0.086 | 0.037 |
| T0010     | ABL1(Q252H)p | 1167   | Yes          | 0.039 | 0.064 |
| T0011     | ABL1(T315I)  | 1167   | Yes          | 21    | 890   |
| T0012     | ABL1(T315I)p | 1167   | Yes          | 3.6   | 120   |
| T0013     | ABL1(Y253F)  | 1167   | Yes          | 0.036 | 0.058 |
| T0014     | ABL1         | 1167   | Yes          | 0.12  | 0.029 |
| T0015     | ABL1p        | 1167   | Yes          | 0.057 | 0.046 |

Table S10: Saved non-mutant EGFR–D0014 functional-group attribution values.

| Functional group | Attribution |
|------------------|-------------|
| Carbonyl         | 0.431       |
| Amide            | 0.389       |
| Ether            | 0.136       |
| Halogen          | 0.134       |
| Amino            | 0.090       |
| Aromatic ring    | 0.000       |
| Heterocycle N    | 0.000       |

Table S11: Saved non-mutant BRAF–D0023 failure-mode summary. The archived affinity is positive at the study threshold, whereas the classifier score is low. The score and attributions are model outputs, not an explanation of binding.

| Quantity                     | Value |
|------------------------------|-------|
| High-affinity-class score    | 0.099 |
| Archived Davis affinity (nM) | 0.19  |
| Aromatic-ring attribution    | 0.059 |
| Hydroxyl attribution         | 0.000 |
| Heterocycle-N attribution    | 0.000 |

Table S12: Trainable parameter count recovered from `interpretability_report.json`.

| Quantity                   | Value     |
|----------------------------|-----------|
| Total trainable parameters | 2,442,083 |

## Strict external BindingDB evaluation

Table S13 records the frozen external evaluation requested by the reviewers. The official curated-articles snapshot was handled independently of Davis, then filtered under the pre-specified exact- $K_d$  and exact double-novel protocol. This is an external exact-entity test, not a claim of scaffold or remote-homology novelty, and no BindingDB label was used to retrain, calibrate, select the checkpoint, or select the threshold.

Table S13: Strict BindingDB external-validation provenance, curation, and frozen result. The source archive was `BindingDB_BindingDB_Articles_202607_tsv.zip` (SHA-256 `d2584d1519318d00ab5f46289da5ab3549affe732d598a5072f8777b6b3b5262`). Only finite exact  $K_d$  measurements in nM for valid single-chain proteins (maximum 1,200 residues) and valid ligand structures were considered. Exact canonical-isomeric-SMILES and full-normalised-sequence matches to every Davis ligand or target were excluded. Eligible measurements were grouped by canonical ligand-sequence pair; threshold-conflicting groups were removed at the pair level, and every remaining pair retained its median exact  $K_d$ . Confidence intervals are pair-stratified bootstrap 95% intervals from 600 replicates (seed 42).

| Item                                             | Value                                                                                                                                            |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Source snapshot                                  | <code>BindingDB_BindingDB_Articles_202607</code> (official BindingDB curated articles TSV)                                                       |
| Source rows                                      | 93,712                                                                                                                                           |
| Final positives / negatives                      | 516 / 1,416                                                                                                                                      |
| Unique ligands / targets                         | 1,195 / 396                                                                                                                                      |
| Sequential row-level exclusions                  | 82,988 censored/non-exact $K_d$ ; 8,261 multi-chain; 139 invalid sequences; 8 invalid SMILES; 64 Davis-ligand matches; 94 Davis-sequence matches |
| Eligible measurements before pair reconciliation | 2,158                                                                                                                                            |
| Threshold-conflicting replicate groups           | 15 ligand-target pairs, comprising 44 measurement rows; conflicts are counted at the pair level                                                  |
| Threshold-consistent measurement rows            | 2,114                                                                                                                                            |
| Median replicate aggregation                     | 143 retained pairs had $\geq 2$ consistent measurements; 182 repeated measurement rows were collapsed by the within-pair median $K_d$            |
| Final external pairs                             | 1,932                                                                                                                                            |
| Frozen model and threshold                       | <code>checkpoints/best_random.pt</code> ; 0.5959881544 (existing Davis random-validation threshold)                                              |
| Protein features                                 | Newly extracted <code>fair-esm 2.0.0 esm2_t30_150M_UR50D</code> ; all 396 caches passed 640-dimensional and length checks                        |
| AUROC (95% CI)                                   | 0.5597 [0.5313, 0.5885]                                                                                                                          |
| AUPRC (95% CI)                                   | 0.3404 [0.3138, 0.3724]                                                                                                                          |
| Frozen-threshold F1 / precision / recall         | 0.0701 / 0.7308 / 0.0368                                                                                                                         |
| Two-run reproducibility                          | Identical per-pair probabilities; the complete SHA-256 is recorded in <code>bindingdb_external_manifest.json</code>                              |
| Interpretation                                   | Limited external transfer; the result does not support a broad DTI-transfer claim and is not used for structural or attribution interpretation.  |